

Spatio-Temporal Association Rule based Deep Annotation-free Clustering (STAR-DAC) for Unsupervised Person Re-identification

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Abstract

Multi-camera video surveillance environment has a variety of emerging research problems among, which person re-identification is the premier one. Unsupervised person re-identification has been explored less in literature than the supervised approach. Images acquired from the video surveillance systems are unlabeled, which denotes that it is naturally an unsupervised learning problem. The state-of-the-art unsupervised methods seek external annotations support such as incorporating transfer learning techniques, partial labeling of train images, etc., which makes them not purely unsupervised and unsuitable for practical real-world surveillance settings. Identity mismatch happens due to the similar costumes and complex environmental factors. To resolve this issue, we introduce a new framework named Spatio-Temporal Association Rule based Deep Annotation-free Clustering (STAR-DAC) which incrementally clusters the unlabeled person re-identification images based on visual features and performs cluster fine-tuning through the mined spatio-temporal association rules. STAR formulations leveraged upto 75% of images for reliable sample selection through cluster fine-tuning. STAR based fine-tune algorithm aims to attain ground-truth labels of an unlabeled dataset and eliminate cluster outliers to stabilize the evaluation. Experiments are performed on image and video-based benchmark person re-identification datasets such as DukeMTMC re-ID, Market1501, MSMT17, CUHK03, GRID and Dukevideo re-ID, iLIDSvid,

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ViPer respectively. Experimental results clearly show that the proposed STAR-DAC framework outperforms the state-of-the-art methods in case of large scale datasets with multiple cameras.

Keywords: Unsupervised Person Re-identification, Clustering, Labeling, Spatio-temporal, Deep Learning

1. Introduction

Person re-identification (re-ID) task is to match the person images of same identities captured from multiple non-overlapping cameras. The training data must be labeled close to its ground-truth labels to efficiently learn the salient features of a person [1, 2]. Traditional benchmark datasets used in the state-of-the-art person re-identification methods are hand-craft labeled. In a real-world multi-camera surveillance environment, the person re-ID images captured are naturally unlabeled (i.e., new people entering into the environment are unknown). Thus, the supervised and semi-supervised person re-ID methods fail in a real-time environment. This motivates to propose a practical unsupervised approach that labels the unlabeled person re-ID images efficiently and improve matching accuracy.

Unsupervised learning techniques are adopted to label the unlabeled person re-ID images through clustering. In person re-identification, unsupervised approach suits the real-time and practical surveillance settings than the supervised approaches. The unsupervised person re-identification methods depend on pre-trained models and transfer learning techniques. In this work, we explore the unsupervised person re-identification problem without using external annotations.

Annotations can be defined into two categories namely with external annotation and without external annotation. With external annotations denote the presence of factors such as supervised learning models, pre-trained model initialization with a labeled source dataset, transfer learning, initializing models with partially labeled datasets, etc., which is not purely unsupervised and unsuitable for practical real-world video surveillance settings. Without external annotation support denotes the methods which involve only auxiliary dataset details like timestamp, camera id, camera start-time and frame id. This information can be easily collected from the actual video surveillance system. Such an approach suits to be a more practical, incremental and self-learning.

Hierarchical clustering techniques are used to cluster the unlabeled person re-ID images. A valid set of parameters, metrics and association rules are required to fine-tune the clusters. The challenging task is to find a cluster convergence point at which the number of clusters is equal to original ground-truth identities. The benchmark person re-ID datasets provide the ground-truth labels which are removed to perform the clustering and fine-tune process. However, the ground-truth labels are used to evaluate the proposed model for benchmarking the obtained results with the existing person re-ID methods. Visual similarity based feature clustering fails due to occluded or illuminated images and similar appearance of different identities. Instead of having no cues about such scenarios, deriving strategies for appearance prediction and outlier elimination improve the performance of unsupervised person re-identification methods.

The benchmark person re-ID datasets are pre-processed and contain only visual features extracted from bounding box images. So, there is a need to backtrack deeper into the neglected auxiliary informations such as camera id, timestamp, frame id, etc., to unravel the underlying new spatio-temporal patterns [3]. In a practical multi-camera surveillance environment, spatial (camera id) and temporal (timestamp) information can be acquired from the actual surveillance videos. These spatio-temporal information is mined to formulate association rules for visual cluster fine-tuning which improves unsupervised person re-ID performance. In supervised person re-ID, few spatio-temporal methods [4, 5] exist with less exploration. In unsupervised person re-ID, spatio-temporal pattern based visual cluster fine-tuning has not been explored and STAR-DAC is the first such approach to the best of our knowledge.

Spatio-temporal cluster fine-tune resolve the issues caused by occluded or illuminated images and person with similar costumes as shown in Figure 1.

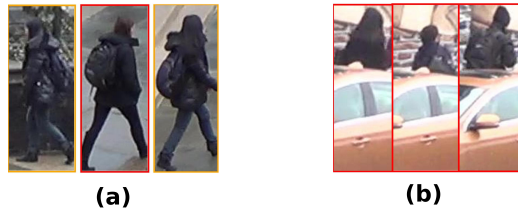


Figure 1: (a) Sample identity mismatch scenario due to similar appearance, (b) Occluded noisy ground-truth samples from DukeMTMC re-ID dataset

Ground-truth images denote the images which are originally labeled to its actual identity. Noisy images denote the occluded low quality images. Thus, the ground-truth images which are not reliable and occluded are called noisy ground-truth images. The spatio-temporal patterns obtained from within-camera and cross-camera relationships help to overcome the visual similarity issues through outlier elimination.

The contributions of this work are summarized as follows:

1. We propose a novel unsupervised deep learning framework named Spatio-Temporal Association Rule-based Deep Annotation-free Clustering (STAR-DAC) which performs visual cluster fine-tune to label the unlabeled person re-ID images without external annotations support.
2. We propose a Spatio-Temporal Association Rules (STAR) formation algorithm to discover the underlying spatio-temporal patterns such as camera correlation, intra-cluster and inter-cluster relationships.
3. The proposed STAR fine-tune algorithm reduce the sample loss by 30 % during reliable sample selection.
4. Experimental results show that STAR-DAC solves the unsupervised person re-ID problem with a maximum improvement of +10.1 % rank-1 and +4.6 % mean Average Precision (mAP) accuracy (evaluated on DukeMTMC re-ID) than the state-of-the-art methods.

The rest of the paper is organized as follows. Section 2 discusses the state-of-the-art unsupervised person re-ID and spatio-temporal methods. Section 3 presents the proposed STAR-DAC framework. Section 4 discusses about the datasets, experimental setup and analyses the experimental results in comparison with the existing methods. Section 5 concludes the work with future directions.

2. Comparative literature analysis of deep unsupervised person re-identification methods

Deep learning based person re-ID methods are categorized into supervised, semi-supervised and unsupervised learning problems. Unsupervised person re-ID methods based on visual feature clustering and different variants of spatio-temporal models are discussed.

2.1. Visual feature based unsupervised person re-identification

The unsupervised person re-ID methods are further categorized based on the type of datasets as image-based and video-based person re-ID methods.

2.1.1. Image-based person re-identification methods

95 Peng et al.[6] claims to be unsupervised by imparting unlabeled images as target dataset and labeled images as source dataset. The feature knowledge acquired from source dataset is transferred to the target unlabeled dataset which indicates that the model is not purely unsupervised and incremental. Wei et al.[7] introduced a new dataset named MSMT17 (Multi-Scene Multi-
100 Time) which contains images under different scenarios such as lighting changes, background scene changes, pose variations and occlusions. By training a model on MSMT17 and testing on different dataset helps to learn complex feature variations. They performed various experiments by transferring domain knowledge from large-to-small and large-to-large datasets with GoogleNet [8]
105 as the backbone neural network.

Fan et al.[9] performed reliable sample selection at each level of clustering. The reliable sample selection is based on visual similarity between images which eliminates the images with a different pose, occlusion and illumination. Eliminating the noisy samples in each iteration leads to a 60 % loss of images
110 in a dataset. More number of noisy samples per identity makes the model vulnerable to the loss of complete identity. They report that using more than one source labeled dataset for training initialization causes a drop in performance and unsuitable for all type of scenarios.

Lu et al.[10] incorporates a centralized clustering loss function which
115 rectifies the issue faced by the clusters due to the softmax layers. Evaluation results show better clustering quality with [9] as the base model. This model has not rectified the need of an additional model trained by a labeled dataset. Xian et al.[11] uses multiple convolutional neural networks initialized with different labeled datasets. Further, they are fine-tuned by the soft labels
120 formed during clustering and exchange process. The model needs many labeled source datasets for initialization and improves the transfer learning process which again makes it not purely unsupervised.

Zhang et al.[12] imparts fuzzy labeling mechanism through similarity and membership degree between samples and clusters. The fuzzy clustering
125 algorithms result in clusters where a sample might belongs to more than one cluster which is impractical for person re-ID problems. Su et al.[13] proposed a semi-supervised approach which learns the mid-level robust visual attributes through small set of labeled images. Cross dataset fine-tuning is performed at the target domain based on triplet loss. Song et al.[14] proposed
130 a generalizable re-ID meta learning model to learn the mapping between

image and its identity classifier in source domain without influencing the target domain. Jia et al.[15] presents a domain generalization model based on the instance and feature normalization applied in source domain and transferred to target domain. Zhao et al.[16] proposed a supervised learning based joint transfer learning technique which combines multiple views and parts of the body to perform re-ID task. Xiang et al.[17] proposed a re-ID system by applying Long Short-Term Memory (LSTM) to each body segment. Raj et al.[18] proposed a person re-ID method which segregates the multi-view pedestrian poses through a representation model.

2.1.2. Video-based person re-identification methods

Unsupervised video-based person re-ID methods involves the surveillance videos which are generated as image tracklets. These tracklets are used as input images by the video-based methods. Li et al.[19] incrementally discovers the salient features of tracklets generated from surveillance videos. Zheng et al.[20] extends [19] and performs association strategies between the tracklets of same and different identities. The tracklets are generated from Dukevideo re-ID dataset at a rate of 65.9 frames/sample. Wang et al.[21] transfers discriminative features to the unlabeled target domain. It uses supervised learning approach in source domain to learn the attribute semantics. Yu et al.[22] learns the specific view based features using asymmetric clustering. This improves performance through a common feature space in case of different view angles.

Ma et al.[23] presents a similar cross-view alignment based video matching method without pairwise labeled images. Liu et al.[24] presents a step-wise metric promotion method which uses each training tracklets as a query image to evaluate the retrieval accuracy with remaining camera images as gallery set. Ye et al.[25] estimate the labels at intermediate levels for feature learning and frequent image graph update through feature space representation. Bak et al.[26] introduces a Synthetic Person Re-Identification (SyRI) database to overcome illumination ambiguity in the unseen target domain. Chen et al.[27] presents a joint matching model based on end-to-end optimization with cross-camera rank list evaluation. Frame wise association is similar to the tracklet association method [20].

2.2. Spatio-temporal pattern based person re-identification

Spatio-temporal information in the video surveillance environment can be used to improve the performance of person re-ID methods [28]. Wang et

al.[4] transfers visual features from a labeled source dataset to target domain aided by spatio-temporal patterns. Lv et al.[29] incorporates bayesian fusion to combine visual and spatio-temporal model for performance evaluation. 170 Martinel et al.[30] proposed a graph-based supervised approach which selects the most informative prob-gallery matches. Cho et al.[5] jointly gives solution to person re-ID and camera topology inference. The above discussed spatio-temporal methods are supervised approaches where the unsupervised approach is discussed in the proposed work.

175 Limitations of the existing methods are summarized as,

- The methods adopt a labeled source train dataset which does not guarantee performance improvement for all scenarios and makes the evaluation protocol unreliable.
- The reliable sample selection removes nearly 60 % of images which 180 causes huge information loss. Thus, the clustering mechanism can never label the datasets to its complete ground-truth labels due to insufficient images. This possess a vulnerability to lose a complete noisy identity. which leads to misclassification.
- The methods [9, 10, 11, 12] (discussed in section 2.1.1) incorporate k- 185 means clustering demands the number of clusters to be initialized before starting the clustering process. To overcome this, the proposed method performs hierarchical clustering without prior cluster initialization.
- Fan et al.[9] reported that many labeled source dataset initialization degrades the person re-ID performance. Thus, different source dataset 190 initialization does not guarantee a performance improvement in all types of scenarios.
- Incorporating spatio-temporal model into the unsupervised person re-ID setup has not been explored.

195 The discussion infers that visual feature based methods are uncertain in an unseen complex surveillance environment. Thus, there is a need to explore the strategies which unravel non-visual cues underlying in the metadata information of surveillance system and dataset. Detailed explanation of proposed method and formulations to resolve the limitations are discussed in subsequent sections.

200 3. Proposed STAR-DAC Framework

To overcome the limitations of existing unsupervised person re-ID methods discussed in the previous section, we propose a novel framework named Spatio-Temporal Association Rule-based Deep Annotation-free Clustering (STAR-DAC) which labels the unlabeled person re-ID images without any
205 external annotations support. The proposed spatio-temporal association rule (STAR) formation algorithm discovers the underlying spatio-temporal patterns and camera correlation of the visual clusters. The mined association rules are used to fine-tune the visual clusters which lead into three broad benefits,

- 210 • Improves the labeling process to obtain ground-truth labels of a dataset.
- Reduces the sample loss during reliable sample selection by 30 % through STAR fine-tune.
- Resolves the ambiguity created by different identities wearing similar costumes.

215 These benefits are obtained by the formulations and algorithms proposed for visual cluster fine-tuning. To the best of our knowledge, the proposed STAR-DAC framework is the first to label the unlabeled person re-ID images without external annotations support and by effectively utilizing the auxiliary spatio-temporal information. Figure 2 shows an overview of the proposed
220 STAR-DAC framework and Algorithm 1 explains the steps involved in the process flow. The architecture is divided into two broad modules namely visual feature clustering and spatio-temporal association rule formation.

3.1. Visual Feature Clustering

225 Unlabeled person re-ID images acquired from the surveillance system are fed to the convolutional neural network for visual feature extraction. The state-of-the-art methods which incorporates k-means clustering requires number of cluster initialization to estimate initial centroids. In the real-world scenario, the exact number of identities in an unseen video surveillance environment is unknown.

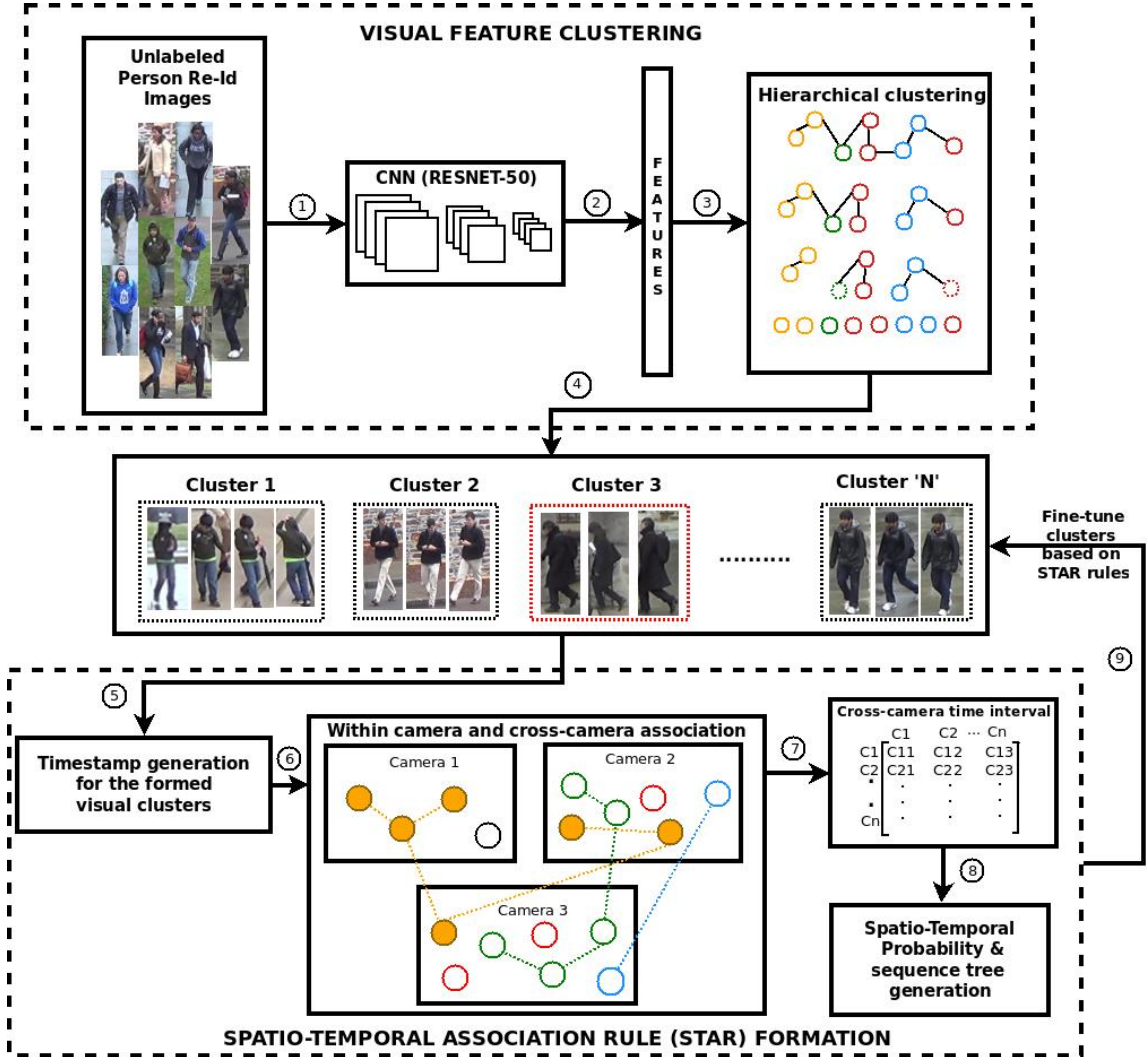


Figure 2 Overview of STAR-DAC framework. The process flow indicated with numbers in circle are as follows, 1) unlabeled images fed to CNN, 2) feature extraction, 3) perform hierarchical clustering on the features, 4) clusters formed, 5) cluster wise timestamp generation from frame numbers, 6) clusters are processed to mine within-camera and cross-camera association, 7) generates cross-camera time interval matrix for mined values, 8) generates spatio-temporal sequence and probability tree and 9) cluster fine-tuning until convergence. (red color dotted box and circles denote sample instance of different identities wrongly merged into a cluster)

Algorithm 1 STAR-DAC Framework

```
1: procedure STAR BASED VISUAL CLUSTER FINE-TUNE
2:   Input: Unlabeled images  $X = [x_1, x_2, x_3, \dots, x_n]$ , Camera_id  $D = [d_1, d_2, d_3, \dots, d_l]$  where  $l$  is the total number of cameras, Timestamp  $T = [t_1, t_2, t_3, \dots, t_n]$  where  $n$  is the total images with  $X = T$  //each image has a timestamp
3:   Output: Fine-tuned visual clusters based on STAR formulations
4:   Initialize: Cluster labels  $C = [C_1, C_2, \dots, C_n]$ , No of clusters  $C = X$ , ResNet-50 as backbone neural network //Each unlabeled image is initialized as an individual cluster
5:   for each  $x_i \in X$  do
6:     Extract features  $\leftarrow$  features of all  $X$  images
7:   end for
8:   while (!cluster convergence) do // Converges for STAR formulations
9:     clusters  $\leftarrow$  Agglomerative_clustering(features)
10:    Perform STAR formation Algorithm 2
11:    Perform STAR cluster fine-tune Algorithm 3
12:     $C \leftarrow$  Update with the fine-tuned cluster labels
13:    Re-train CNN with fine-tuned clusters in  $C$ 
14:   end while
15: end procedure
```

230 To overcome the cluster initialization ambiguity, number of clusters are initialized to number of unlabeled images in the dataset which is as similar as not defining the clusters. Agglomerative hierarchical clustering is performed on the normalized visual features with regularization. Clustering process is evaluated for different distance threshold values with cosine similarity as the distance measure. Rank accuracy and mean average precision (mAP) 235 values are used to observe the cluster convergence with less intra-cluster and large inter-class variations. The finalized clusters are forwarded to the Spatio-Temporal Association Rules (STAR) formation module to discover the underlying spatio-temporal patterns and camera correlation.

240 *3.2. Spatio-Temporal Association Rule (STAR) formation*

Visual clusters are fed to the STAR formation module (Algorithm 2) along with the auxiliary spatio-temporal details (camera id, timestamp, camera start-end time, frame numbers and number of frames per second). These

245 auxiliary details are used to generate timestamp for the samples. The intra-cluster samples are sorted based on the spatial (camera id) and temporal (timestamp) data to form the spatio-temporal sequence.

Algorithm 2 Spatio-Temporal Association Rule (STAR) Formation Algorithm

```

1: procedure STAR PATTERN FORMATION
2:   Input: Clusters  $C = [C_1, C_2, C_3, \dots, C_f]$  where  $f$  denotes total clusters
      formed at first iteration, Clustered images  $I = [I_1^{C_i}, I_2^{C_i}, I_3^{C_i}, \dots, I_n^{C_i}]$  where
       $C_i \in C$ , Camera id  $D = [d_1, d_2, d_3, \dots, d_l]$  where  $l$  is the total number of
      cameras, Timestamp  $T = (t_{I_j}^{C_i})$  where  $j \leftarrow$  no. of images in cluster  $C_i$ 
3:   Output: STAR formulations with cross-camera matrix and sequence
      tree
4:   for each  $c_i \in C$  do
5:     Sort  $(T_{I_j}^{C_i})$  //Sort images based on timestamp
6:      $\text{adj\_Matrix}_i \leftarrow$  Camera adjacency matrix
7:     Formulate Matrix  $M_i \leftarrow$  camera pairwise timestamp difference
8:     Generate intra-cluster spatio-temporal sequence tree for  $M_i$ 
9:     Matrices  $M \leftarrow M_i$ 
10:  end for
11:  Similarity( $M$ )  $\leftarrow$  Calculate co-relation  $\forall M_i \in \mathcal{M}$ 
12:   $L \leftarrow$  Sort(Similarity( $M$ )) //construct lookup table
13:  while  $M_i \in L$  do
14:    if  $\text{similarity}(M_i) \geq \lambda$  then
15:       $M\_patterns \leftarrow M_i$  //Updated with frequent patterns
16:      Generate cross-camera time interval and probability tree
17:    else
18:      remove  $M_i$  // Outlier matrix elimination
19:    end if
20:  end while
21: end procedure

```

250 Cross-camera time interval matrix and spatio-temporal sequence tree are generated which can be dynamically updated through cluster fine-tuning. Intra-cluster matrix and sequence tree can be generated for each cluster to infer identity based information and in handling complex query images. To evaluate the cross-camera similarity matrix for the whole environment, a

similarity lookup table L is constructed containing all the clusters. Matrices with similar time interval are clustered to mine the frequent cross-camera patterns. Thus, spatio-temporal pattern formation Algorithm 2 eliminates the outlier matrices to form the stable cross-camera time interval matrix.

3.2.1. STAR formulations to fine-tune clusters

STAR cluster fine-tune Algorithm 3 incorporates the following formulations to deliver the fine-tune visual clusters. The STAR formulations are,

Formulation 1. Consider a pair of cameras m and n . Image I_i appeared in Cam_m at timestamp t has a low probability of appearing in Cam_n within a minimum time interval $\min(\Delta t_{mn})$.

$$\Delta t_{mn} = \sum_{k=1}^Y \sum_{i,j=1}^I t_m(I_i^{c_k}) - t_n(I_j^{c_k}), \quad \forall m, n \quad (1)$$

where $\Delta t_m(I_i^{c_k})$ denotes image $I_i \in k^{th}$ cluster captured in camera m at timestamp t and $\Delta t_n(I_j^{c_k})$ denotes another image $I_j \in k^{th}$ cluster captured in a different camera n at timestamp t . Intra-cluster samples are sorted based on temporal sequence. Then perform equation 1 for all image and camera pairs. The outlier matrices are eliminated with a threshold value λ initialized to 1 (adjustable as per the environmental factors) which leads to the cross-camera time interval matrix and sequence tree.

Formulation 2. Within-camera time interval is derived by formulating the timestamp difference between the sorted image pairs of every cluster. The time interval depends on the number of images captured by a camera in sequence or with certain intervals.

$$\Delta t_{Cam_m} = |I_i^m - I_{i+1}^m|, \quad \forall (I_i^m, I_{i+1}^m) \in c_k \quad (2)$$

where Δt_{Cam_m} is the within-camera time interval difference within a cluster after outlier elimination. In equation 2, for each pair of $(I_i^m, I_{i+1}^m) \in Cam_m$, the difference is calculated and outlier elimination through mode operation is applied to arrive at the final within-camera time interval value. These values can be used to formulate the time spent by each of the identities in the respective cameras. If the captured images have a high timestamp interval, then it denotes the person often revisits the camera. The person seems to stay in the camera for a long time without moving to the other cameras in case of low time interval between captured images.

275 **Formulation 3.** Two images captured at same timestamp t cannot belong to the same cluster, provided the dataset is free from duplicate or redundant images. $\text{Similarity}(I_i, I_j) > \delta_{\min}$ when $t(I_i) = t(I_j)$, then the images belongs to different identities. Similarity lookup table L_i which contains the cluster identity and similarity score is constructed for both the
280 images. The outlier image is identified and mapped to its corresponding cluster through Algorithm 3.

Formulation 4. The spatio-temporal sequence tree is generated from within-camera and cross-camera time interval matrix. $\text{level}(\text{Cam}_m) = 0$ denotes that Cam_m is the starting point of the sequence. The sequence tree
285 can be generated for different root nodes by initializing value of $m \leq \text{total cameras } D$. Flag variable $\text{visit}(\text{Cam}_m)$ is assigned to each of the cameras. For each visit of camera m , $\text{visit}(\text{Cam}_m)$ is incremented gradually to obtain total count of revisits. This parameter contributes an additional bias factor for the particular camera images during the pattern mining process.

Formulation 5. Cross-camera trajectory movement frequency reduces the computation complexity during the fine-tuning process. Consider the spatio-temporal sequences $S = [S_1, S_2, S_3, \dots, S_Z]$ of corresponding clusters $C = [C_1, C_2, C_3, \dots, C_Z]$ where Z denotes the total number of clusters and its corresponding sequences. S_i^c denotes the sequence of i^{th} cluster $\in C$. For each pair of cameras k and l in the sequence S_i perform the following,

$$\sum_{i=1}^Z \sum_{j=i+1}^Z [S_i^{C_i}(\text{Cam}_k, \text{Cam}_l) + 1, S_j^{C_j}(\text{Cam}_k, \text{Cam}_l) + 1], \quad \forall (S_i^{C_i} = S_j^{C_j}) \quad (3)$$

where $S_i^{C_i}(\text{Cam}_k, \text{Cam}_l)$ denote the movement between each of the camera pairs Cam_k and Cam_l in the sequence $S_i^{C_i}$. Each time the path is visited, $(\text{Cam}_k, \text{Cam}_l)$ is incremented by 1. To update the probability of entire sequence,

$$\sum_{i=1}^Z \sum_{j=1}^Z \text{Pr}(S_i^{C_i}, S_j^{C_j}) + 1, \quad (S_i^{C_i}, S_j^{C_j}) \in S, C \quad (4)$$

290 where $\text{Pr}(S_i^{C_i}, S_j^{C_j})$ is the overall sequence probability of the established trajectory $\forall (\text{Cam}_k, \text{Cam}_l)$ in the sequences S_i and S_j . The probability values calculated by equation 3 and 4 are normalized to a scale of 0 to 1 and used for individual cluster-based spatio-temporal probability tree generation. This helps to reduce the search-space while performing the cluster fine-tune
295 Algorithm 3.

Formulation 6. To mine the predecessor and successor spatial information, (Cam_{m-1}, Cam_{m+1}) is formulated to derive the most probable Cam_{m-1} and Cam_{m+1} for a given Cam_m . This can be extended for any number of predecessor and successor cameras.

300 **Formulation 7.** To mine the entire sequence tree pattern, from the root camera to last level cameras is traversed. This spatio-temporal sequence is used to derive the most probable sequence which acts as an additional factor in leveraging the similarity threshold from δ to δ_{min} during the cluster fine-tune process.

305 As new images are added to the gallery, the formulations dynamically update the values based on the mined association rules which makes the algorithm purely incremental and self-learning.

3.2.2. Spatio-temporal cross-camera probability tree formation

STAR formation Algorithm 2 also generates a spatio-temporal probability matrix which represents the occurrence frequency of within and cross-camera sequence. It also shows the probability of a person's appearance in subsequent cameras derived from the formulations 5,6 and 7. Different probability trees can be constructed by initializing each camera as the root node for each cluster. Spatio-temporal tree generation ends when the number of levels is equal to total number of cameras. Figure 3 shows the cross-camera movement frequency of DukeMTMC re-ID dataset.

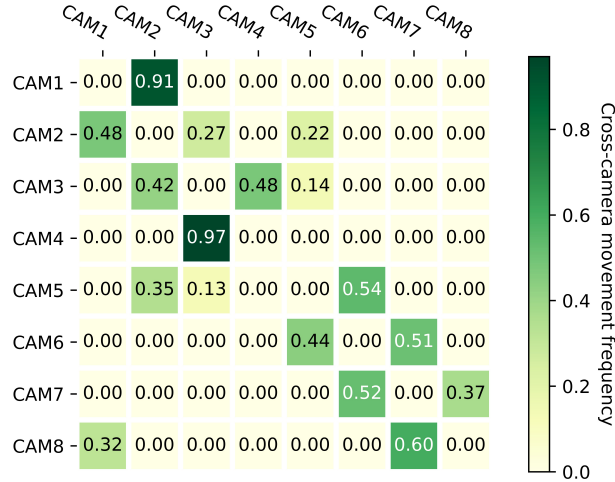


Figure 3: Cross-camera movement frequency matrix for DukeMTMC re-ID

It generates a $m \times m$ matrix where m is the number of cameras in the environment. The camera pairs with value 0 denote no movement recorded. This narrows down the fine-tuning process by restricting the camera evaluation only to the movement existing pairs. The probability value mentioned from 0 to 1 denotes low to high movement frequency.

Cam 1 and Cam 4 have the movement only with one other camera which denotes that they are at the extreme ends where the movement is restricted to only one other camera (as shown in Figure 3). This type of patterns resolve the ambiguity when the investigated query image lies in any of these extreme cameras. It adds an additional clue to the query image by predicting the next spatial location the identity should have visited.

Cam 1 being the starting point (root node), the spatio temporal probability sequence tree of DukeMTMC re-ID dataset majorly takes two path sequences. Figure 4 shows the sequence generated based on the maximum probability at each level of the tree. Figure 5 shows the second most probable locations visited which directs to Cam 5 instead of Cam 3 at level 3.

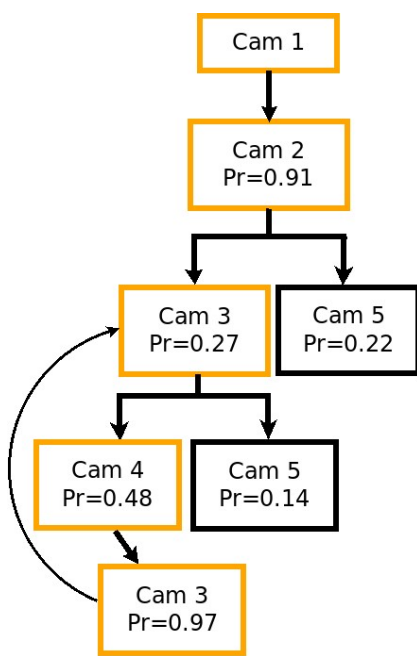


Figure 4: DukeMTMC re-ID sample sequence 1

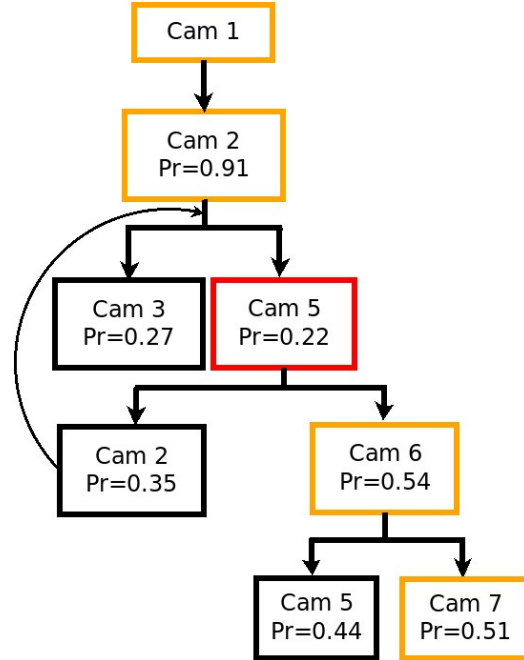


Figure 5: DukeMTMC re-ID sample sequence 2

Depending upon the sequences formed by the visual clusters, cross-camera and within-camera sequence can be matched with their corresponding sequences to fine-tune the clusters through outlier elimination.

3.3. Spatio-temporal Association Rule (STAR) based cluster fine tuning

The proposed spatio-temporal fine-tune Algorithm 3 fine-tunes the visual clusters based on the STAR formulations. Initially, intra-cluster similarity is evaluated among the cluster samples. If the similarity of a image I_i is higher than similarity threshold δ , then formulation 1 is evaluated.

Algorithm 3 Spatio-temporal cluster fine tuning algorithm

```

1: procedure STAR CLUSTER FINE-TUNE
2:   Input: Clusters  $C = [C_1, C_2, C_3, \dots, C_k]$ , Images  $I = [I_1, I_2, I_3, \dots, I_n]$ ,
   any camera pair  $m, n \in C$ , STAR formulations
3:   Output: Fine-tuned clusters
4:   for each  $I_i \in C$  do
5:     repeat
6:       if  $\text{similarity}(I_i^{C_k}) > \delta$  then
7:         // evaluated for formulations 1,2,3 and 4 in Section 3.2
8:         repeat
9:           if  $\Delta t_{mn} > \min(\Delta t_{mn})$  then //  $\forall m, n \in C$ 
10:             $C_k \leftarrow C_{k+1}$  // Verify the next cluster
11:          else
12:            Retrieve image  $I_i^{C_k}$ 
13:            Construct  $L_i$  for the retrieved image  $I_i$ 
14:             $L_{C_i} \leftarrow L_{C_i} + 1$  //  $l$  denotes total clusters in  $L_i$ 
15:            //Evaluate for the next clusters
16:          end if
17:        until  $\text{similarity}(L_{C_i}) \geq \delta_{min}$ 
18:      end if
19:       $\delta \leftarrow \delta - 0.05$ 
20:    until  $\delta \geq \delta_{min}$ 
21:    Outlier Image (O)  $\leftarrow I_i^{C_i}$ 
22:    Noisy clusters  $\leftarrow O$ 
23:  end for
24: end procedure

```

STAR formulation conditions has to be satisfied for a cluster to proceed with the evaluation of next cluster. If image $I_i^{c_k}$ contradicts the formulations, then the image is retrieved to map it to the suitable cluster. A similarity lookup table L_i is constructed for the retrieved image I_i where i denotes identity of the image and its corresponding lookup table. The clusters in similarity lookup table L_i are sorted in decreasing order of similarity score. Image I_i is verified to the formulations with each of the cluster $L_{c_i} \in$ Lookup table L_i . The cluster fine-tune continues until similarity value of image I_i reaches minimum threshold value δ_{min} .

The cluster similarity δ is leveraged by 0.05 to allow further images from the dataset for performing STAR cluster fine-tune procedure. As the images increase, spatio-temporal patterns become stable gradually. Until the STAR rules fail, clusters are cross verified with the subsequent clusters in the lookup table upto δ_{min} value below which the fine-tune faces a performance drop due to noisy images. But, on an average, 30 % extra images are allowed by the proposed cluster fine-tuning Algorithm 3.

The noisy images whose similarity is lesser than δ_{min} are clustered at the end of lookup table L . This denotes that there are more clusters than the exact ground-truth identities which will contain the noisy images. These noisy images are not considered for the visual feature formation or spatio-temporal pattern mining. The noisy images are added in order to allow the threshold value to drop till 0 by which complete ground-truth labels are obtained. The noisy images affects the feature of a person, thus the similarity limit is not leveraged beyond a threshold (i.e., <0.75).

3.4. Rank level STAR based retrieval/fine-tune of clusters

The query-based ranking evaluation method also follow the same strategy as Algorithm 3. Consider a query set $Q = [Q_1, Q_2, Q_3, \dots Q_n]$, Gallery or test set $G = [G_1, G_2, G_3, \dots G_n]$. The gallery set contains ground truth labels of images which are not in the training set. Train and test split details for all the datasets are given in Table 1. Initially, $\forall Q_i \in Q$, visual features are extracted and evaluated with similarity of the retrieved gallery images.

Consider a user querying gallery G with a query image Q_i . Let $d(Q_i, G_j)$ denote the similarity between Q_i and G_j . The gallery set is ordered as G' according to $d(Q_i, G_j)$ score of the query image Q_i . $d(Q, G)$ denote the cumulative similarity scores for all queries. Using G' and the ground-truth labels, precision, recall and average precision values are calculated. Rank accuracy has been computed for rank-1, 5, 10 and 20 with mAP (mean

Average Precision) which represents the Cumulative Match Characteristic(CMC) scores. The image retrieved to its respective class after passing a large
 380 number of ranks leads to a low average precision. This denote that precision decreases as the rank increase for a given query Q_i .

Initially, if similarity $d(Q_i, G_j) > \delta$, then it is considered for CMC score evaluation. All the query-gallery image pairs which satisfy the similarity conditions are retrieved. STAR formulations are used to map the outlier
 385 images to its corresponding class. The rank evaluation strategy leverages the threshold value of δ upto δ_{min} (derived based on empirical study shown in Figure 9 and 12) to match a image to its correct ground-truth class. The threshold leverage is performed by taking the advantage of STAR fine-tune Algorithm 3 which improves the overall mAP.

Gallery images G_i whose $d(Q_i, G_j) < \delta$ for all the evaluated queries are
 390 left as unreliable visual mismatches. These images are considered to be noisy images which affect the mean average precision. STAR fine-tune strategy of Algorithm 3 is applied to the rank evaluation process to match the retrieved image to their corresponding classes. At the end of STAR fine-tune, the
 395 matched rank accuracy and mAP is evaluated.

$$AP = 1/GTP \sum_{i=1}^K TP/i \quad (5)$$

where AP - Average Precision, GTP - Ground Truth Positives, TP - True Positives. Equation 5 computes the average precision for each query match and mismatch with with the ground truth positives. mAP is evaluated by the cumulative values of the AP for all the queries in a given dataset.

400 The rank evaluation strategy eliminates and fine-tunes the false positives to its corresponding class which leads to a further increase in mAP accuracy. The accuracy improvement proves that the modified rank evaluation procedure by the strategy of Algorithm 3 matches the retrieved images to its ground-truth labels better than the traditional mAP rank evaluation procedure.

4. Datasets and Experimental result analysis

The proposed STAR-DAC framework has been evaluated on both image-based and video-based person re-ID datasets. The details of the datasets are shown in Table 1.

Table 1: Image and video-based person re-ID dataset details

Datasets	#Cams	#Identities	#Train	#Test	#Images
Image based datasets					
DukeMTMC re-ID [31]	8	1,812	702	1,110	36,411
Market-1501 [32]	6	1,501	751	750	32,668
MSMT17 [7]	15	4,101	1,041	3,060	126,411
RPI field [33]	12	112	55	56	601,581
CUHK03 [34]	2	1,467	767	700	14,097
GRID [35]	8	1,025	500	525	1,275
Video based datasets					
Dukevideo re-ID[31]	8	1812	702	1110	445,764
iLIDSvid [36]	2	300	150	150	43,800
ViPeR [37]	2	632	300	332	1,264

DukeMTMC re-ID (Duke Multi-Target Multi-Camera)[31] contains many persons wearing similar color dresses which make the visual similarity-based clustering process difficult. The frame rate at which the images are captured is 60 frames/second. DukeMTMC re-ID contains the required auxiliary information (frame numbers, frame-level ground truth, camera ID's, etc.). Market-1501 [32] contains 6 cameras captured at 25 frames/second with one low-resolution camera. MARS is an extension of the Market1501, but the frame numbers are not provided which make us impossible to generate timestamps. It also includes 2793 distractor images to resemble the practical environment. MSMT17 dataset [7] contains 15 cameras with 12 outdoor and 3 indoor cameras. It covers 4 days with each day having 3 hours of video with each hour from morning, noon and evening. RPI (Rensselaer Polytechnic Institute) field [33] is the largest person re-ID dataset till date with viewpoints similar to Market-1501. It provides timestamp and revisit frequency in a complex scenario which facilitates the evaluation in a temporal aspect. CUHK03 [34] contains detected bounding boxes which are hand-craft labeled with images captured from 2 disjoint cameras. GRID (underGround Re-ID) [35] contains 2 images per identity collected from 8 non-overlapping cameras in a busy underground station with a very poor image quality. Dukevideo

re-ID [31] consists of 2,196 training videos and 2,636 testing videos captured at every 12 frames. Each identity acts as a query with the remaining videos placed in the gallery. iLIDS Vid (imagery Library for Intelligent Detection Systems)[36] dataset contains 600 trajectories of 300 identities with highly occluded images. ViPer (Visual Person detection made reliable) [37] provides the view angle of 2 cameras containing one image per identity.

Experimental setup. To evaluate the models, CMC scores with mean average precision (mAP) are used. ResNet-50 [38] acts as the backbone neural network with a dropout rate of 0.5. The images are auto-rotated and resized to 224 x 224 pixels. Batch size is 16, stochastic gradient descent is set with a momentum of 0.9 and learning rate is 0.001. The experiments are performed using Python 3.6.8 on Ubuntu 16.04 LTS version. Hardware comprises of 2 x Intel Xeon E5-2640 v4 Processor(20 Core/2.4GHz/25M Cache), 2 x 16GB ECC DDR4 213 MHz RAM and NVIDIA Quadro P5000 16GB GDDR5X graphics card.

4.1. Results and analysis in comparison with the state-of-the-art methods

The proposed STAR-DAC framework has been evaluated using the datasets mentioned in Table1. Visual-DAC denotes visual clustering results without STAR fine-tune and STAR-DAC denotes the results after performing STAR rule-based fine-tuning. Rank-1,5,10,20 accuracy and mAP values for Visual-DAC and STAR-DAC are reported. DukeMTMC re-ID acts as the source dataset for all the dataset evaluation. Since STAR-DAC framework performs clustering and fine-tuning incrementally in the target domain, influence of source dataset to the accuracy is very negligible. Thus, STAR-DAC reports a stable output irrespective of the source dataset. The evaluated datasets are categorized into large-scale and small-scale datasets. Table 2 show the results of STAR-DAC in comparison with the existing methods using rank and mAP accuracy. The results are reported in % which represents the recognition rate. It denotes the performance of re-ID system in identifying the queried images from the gallery set. For example., if mAP is 31.6, then it denotes that the person re-ID system has retrieved 31.6 % of the total number of queries evaluated.

Table 2: Unsupervised Person re-ID Accuracy on DukeMTMC re-ID and Market1501 datasets

Methods	DukeMTMC re-ID					Market1501				
	Rank-1	Rank-5	Rank-10	Rank-20	mAP	Rank-1	Rank-5	Rank-10	Rank-20	mAP
UMDL [6]	18.5	31.4	37.6	44.7	7.3	34.5	52.6	59.6	68	12.4
PTGAN [7]	27.4	-	50.7	-	-	38.6	-	66.1	-	-
PUL [9]	30.4	44.5	50.7	56	16.4	45.5	60.7	66.7	72.6	20.5
CCLF-PUL+[10]	31.2	47.7	54.1	57.8	17.2	46.1	62.4	68.8	74.2	22.2
PUCL [11]	33.1	46.9	53	61.8	19.5	48.5	63.8	69.9	76.3	22.9
FMC [12]	48	62.3	68.1	-	27.8	63.4	79.5	84.8	-	32.4
Visual-DAC	45.6	60.5	65.9	71.1	26.5	57.8	72.3	77.6	82	29.6
STAR-DAC	56.4	72.1	76.5	81.1	31.6	67	80.6	84.9	88.9	33.9

Figure 6 and 7 show the visual similarity based cluster convergence before cluster fine-tune of DukeMTMC re-ID and Market-1501 respectively. The convergence has been depicted from 10,000 clusters as CMC scores remain stable for the previous cluster segments. Rank-1 accuracy and mAP drops between 828 to 600 and 800 to 600 clusters for DukeMTMC re-ID and Market-1501 respectively. The drop in CMC score denotes that the clusters have converged to the number of ground-truth identities (702 for DukeMTMC re-ID, 750 for Market-1501). After the accuracy drop, further merging of clusters will reduce the clusters less than ground-truth identities which causes a further decrease in matching accuracy.

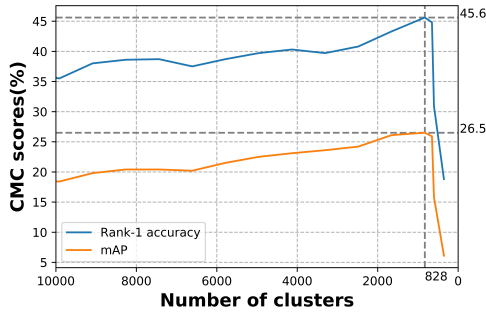


Figure 6: CMC score based convergence before STAR fine-tune for DukeMTMC re-ID dataset

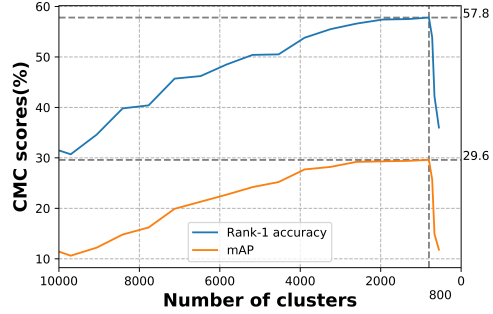


Figure 7: CMC score based convergence before STAR fine-tune for Market1501 dataset

Persons with similar costumes are high (particularly black colored dress) in DukeMTMC re-ID dataset. STAR-DAC performs at its best in such scenarios which results in +8.4 % rank-1 and + 3.8 % accuracy. The dataset provides the exact camera start and end time which is very essential to correlate the cross-camera relationship by which STAR-DAC gives +11.8 % rank-1 and +4.7 % mAP accuracy than the Visual-DAC. Market-1501 shows + 3.6 % rank-1 and +1.5 % accuracy due to one low quality camera. This causes δ_{min} threshold value to conflict with the other five cameras where STAR-DAC (after STAR cluster fine-tune) results +9.2 % rank-1 and +4.3 mAP % accuracy than the Visual-DAC (before STAR fine-tune). The domain features of a pre-trained model influences the clustering process. STAR-DAC fine-tune reduces such influence on the target domain which reports a stable performance irrespective of the source domain. These are two large scale benchmark datasets which are frequently used by researchers in person re-ID.

Table 3 shows that MSMT17 gives a +10.1 rank-1 accuracy and +4.6 mAP. Improvement in mAP is affected by the blurred faces of persons in MSMT17 dataset which causes loss of information in close proximity captures. This ambiguity leads to the mAP decrease of Visual-DAC to -2 % despite the rank-1 accuracy shows a +0.3 % improvement. It has been resolved by STAR-DAC fine-tuning and reports a better accuracy than the existing methods in Table 3.

Table 3: **Unsupervised person re-ID accuracy on MSMT17 datasets**

Methods	MSMT17		
	Rank-1	Rank-10	mAP
PTGAN [7]	18	36.4	6.2
TAUDL [19]	28.4	-	12.5
UTAL [20]	31.4	-	13.1
Visual-DAC	31.7	59.5	11.1
STAR-DAC	41.51	64.86	17.66

Due to large number of images in the RPI field dataset, STAR-DAC shows +7.1 % rank-1 and +6.3 % than the Visual-DAC as shown in Table 6. RPI field being a new dataset, it has no results reported in the literature. The large dataset size and variations in multiple cameras reduces rank accuracy as the rank increases which reflects in the mAP value. This indicates that as the dataset size increases, the clusters with noisy images also increase.

MSMT17 and RPI field contains detailed information about the spatio-temporal information which takes advantage of Algorithm 2 for stable pattern mining. MSMT17 and RPI field spans over 15 and 12 cameras respectively which helps in formulating a robust cross-camera and within-camera patterns to fine-tune the clusters efficiently.

Table 4 shows comparison of proposed method with the literature works on two small scale datasets CUHK03 and GRID. CUHK03 gives a -0.6 % rank-1 and -3.1 % mAP accuracy.

Table 4: **Unsupervised person re-ID accuracy on CUHK03 and GRID datasets**

Methods	CUHK03				
	Rank-1	Rank-5	Rank-10	Rank-20	mAP
UMDL [6]	1.4	5	7.1	11.2	1.9
PUL [9]	8.1	15.8	22	28.5	8.1
CCLF-PUL+ [10]	8.2	14.6	20.7	27.8	9.8
FMC [12]	19.1	25.7	30.1	37.4	18.6
Visual-DAC	16.9	29.9	37.5	45.2	14.7
STAR-DAC	18.5	32.8	39.8	47.4	15.5
	GRID				
	Rank-1	Rank-5	Rank-10	Rank-20	mAP
UMDL [6]	3.97	8.14	10.73	-	-
SSDAL [13]	19.1	35.6	45.8	58.1	-
AGG(DIMN) [14]	29.3	53.3	65.8	-	-
DualNorm [15]	40.5	-	-	-	-
Visual-DAC	34.6	52.8	61.8	69.1	-
STAR-DAC	41.2	57.3	62.2	70.6	-

STAR-DAC shows a +1.6 % rank-1 and +0.8 % mAP than the Visual-DAC. Both Visual-DAC and STAR-DAC give a improved rank-5,10 and 20 accuracy, but a low rank-1 and mAP accuracy. The decrease is due to poor cross-camera patterns with only 2 cameras, though it has 5 views per camera (5x2=10 views). Yet, Algorithm 2 formulates within-camera sequence patterns for fine-tuning. As rank increases, the rank accuracy gradually drops and after a certain rank it drops again resulting in a low mAP. This denotes that, the proposed STAR-DAC takes advantage of the large dataset size in terms of images and number of cameras. GRID dataset results a + 0.7 % increase in rank-1 accuracy. STAR-DAC shows a +6.6 % rank-1 and +7.5 % mAP than Visual-DAC. This denotes that the Algorithm 3 overcomes the disadvantage of poor quality images by cross-camera patterns with 8 cameras. Accuracy improvement than the literature works is low due to less number

of total images, poor image quality and less images per identity. Lowering of
 530 the threshold value below δ_{min} results in a performance drop which indicates
 to stop the cluster fine-tuning as shown in Figure 12.

Table 5 and 6 show the evaluation results on the video-based datasets
 iLIDSvid, Dukevideo reID and ViPer. iLIDSvid gives a +2.5 % rank-1
 accuracy. STAR-DAC gives a +5 % rank-1 and +4.3 % mAP than the
 535 Visual-DAC. iLIDSvid has only 2 cameras for cross-camera patterns which
 is compensated by the within-camera association due to large images per
 identity. Large number of images with less identities slows down the increase
 in rank accuracy which leads to an average mAP. The rank-5,10,20 and mAP
 accuracy are also presented for the proposed method in Table 6.

Table 5: Unsupervised person re-ID accuracy on iLIDSvid (LARGE SCALE), ViPer
 540 (SMALL SCALE) and Dukevideo re-ID (LARGE SCALE) video based datasets

Datasets	iLIDSvid				ViPer		Dukevideo re-ID		
Methods	Rank-1	Rank-5	Rank-10	Rank-20	Methods	Rank-1	Methods	Rank-1	mAP
TAUDL [19]	26.7	51.3	-	82	CAMEL[22]	30.9	TAUDL	61.7	43.5
UTAL [20]	35.1	59	-	83.8	SSDAL[13]	37.9	UTAL	62.3	44.6
UnKiss [39]	35.9	63.3	74.9	83.4	TJ-AIDL[21]	38.5	Visual-DAC	56.6	35.2
DGM-IDE [25]	36.2	62.8	73.6	82.7	DAS [26]	43	STAR-DAC	66.7	46.9
SMP [24]	41.7	66.3	74.1	80.7	Visual-DAC	39.7			
UMDL [6]	49.3	-	-	-	STAR-DAC	40.8			
DAL [27]	56.9	80.6	87.3	91.9					
Visual-DAC	54.4	67.8	74.4	81.2					
STAR-DAC	59.4	74.1	79.5	84.6					

Dukevideo re-ID gives +4.4 % rank-1 accuracy and +2.3 % mAP accuracy.
 STAR-DAC reports a +10.3 % rank-1 and +11.5 % mAP accuracy than the
 545 Visual-DAC due to lesser frame rate between image tracklets of Dukevideo
 re-ID. UTAL [20] and TAUDL [19], report a low accuracy for iLIDSvid
 dataset despite using 833,984 training tracklet images which is 50% greater
 than the size of Dukevideo re-ID dataset. STAR-DAC is able to build the
 model efficiently with only 50% of the tracklets generated with an improved
 550 performance. ViPer gives a -2.2 % rank-1 accuracy due to merely 2 images
 per identity captured from each of the two cameras. This restricts the scope
 of both within and cross-camera pattern mining process reflecting in +1.1 %
 rank-1 and +0.6 % mAP accuracy.

Table 6: Unsupervised Person re-ID matching accuracy of Visual-DAC and STAR-DAC evaluated on all the benchmark person re-ID datasets mentioned in Table 1

Methods	Rank-1	Rank-5	Rank-10	Rank-20	mAP	Rank-1	Rank-5	Rank-10	Rank-20	mAP
	DukeMTMC re-ID					Market1501				
Visual-DAC	45.6	60.5	65.9	71.1	26.5	57.8	72.3	77.6	82	29.6
STAR-DAC	56.4	72.1	76.5	81.1	31.6	67	80.6	84.9	88.9	33.9
	MSMT17					RPI field				
Visual-DAC	31.7	50.9	59.5	68	11.1	43.38	60.7	65.6	71.89	15.39
STAR-DAC	41.51	55.12	64.86	69.55	17.66	50.54	67.38	72.8	77.02	21.65
	CUHK03					GRID				
Visual-DAC	16.9	29.9	37.5	45.2	14.7	34.6	52.8	61.8	69.1	15.2
STAR-DAC	18.5	32.8	39.8	47.4	15.5	41.2	57.3	62.2	70.6	22.9
	Dukevideo re-ID					iLIDSVID				
Visual-DAC	56.6	72.5	78.8	84.1	35.2	54.4	67.8	74.4	81.2	32
STAR-DAC	66.7	80.5	83.5	88.6	46.9	59.4	74.1	79.5	84.6	36.3
	ViPer									
Visual-DAC	39.7	54.2	60.7	66.2	23.6					
STAR-DAC	40.8	55.5	61.4	66.3	24.2					

4.1.1. Reliable Sample Selection Analysis

Figures 8 and 9 show the proportion of reliable images selected for different similarity thresholds. Large scale datasets such as DukeMTMC re-ID, Market-1501, MSMT17 and RPI field allow between 65 % to 72 % of reliable sample images for feature formation and cluster fine-tuning. Small scale datasets like CUHK03 and GRID (denoted in dotted line in Figure 8) allow samples between 50 % to 57 % which is lesser than the large scale datasets. Figures 9 show the proportion of reliable images selected for video-based datasets.

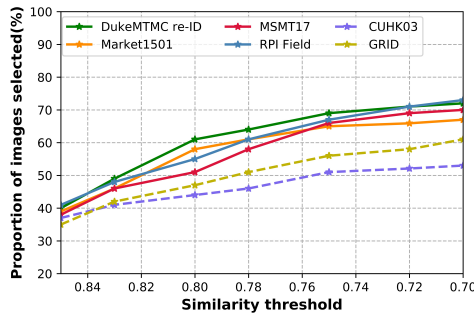


Figure 8: Reliable sample selection during cluster fine-tuning of Image-based datasets

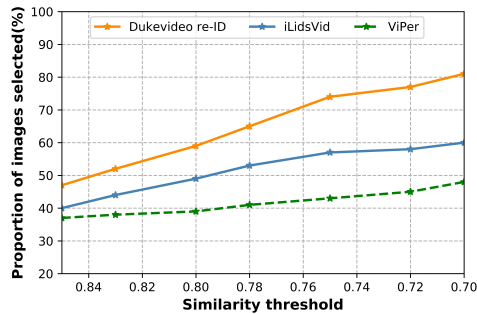


Figure 9: Reliable sample selection during cluster fine-tuning of Video-based datasets

565 Dukevideo re-ID with large number of tracklets, allows upto 75 % of the samples during STAR fine-tune. It is due to the better intra-cluster sample similarity acquired by the tracklets generated in close frame intervals which leads to less noisy images in the dataset. iLIDS Vid dataset allows around 58 % of the sample images for fine-tuning. Despite the large number of images, 570 reliable image selection is directly affected by cross-camera pattern formation with only 2 cameras. ViPer dataset shows a low reliable sample selection of 43 % due to small scale images and camera networks. ViPer being a video-based dataset, it contains only 2 images per identity which denotes the lack of images tracklets. Thus, the dataset takes less advantage of the STAR fine-tune Algorithm 3. The inference from the observation is that, more the 575 tracklets generated from same camera, it allows a high similarity threshold leverage for reliable sample selection.

4.1.2. Cluster quality analysis before and after performing STAR formulation based cluster fine-tune

580 The silhouette score based cluster quality evaluation is performed based on the similarity and dissimilarity of the images which fails to detect the noisy images. Figure 10 shows that the highest silhouette score reported at 828 (DukeMTMC re-ID) and 800 (Market1501) clusters by Visual-DAC (before STAR formulation based fine-tune). STAR-DAC reports highest silhouette score at 927 (DukeMTMC re-ID) and 1014 (Market1501) clusters. 585

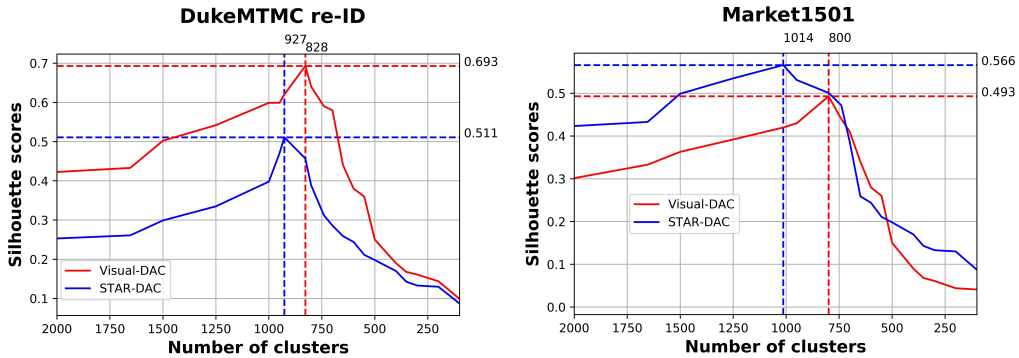


Figure 10: Silhouette score based cluster quality analysis on DukeMTMC re-ID and Market1501 datasets

Visual-DAC reports higher silhouette score at the cluster range closer to actual identities (i.e., 702 and 750 for DukeMTMC re-ID and Market1501

respectively). However, the rank-1 and mAP accuracy of STAR-DAC is higher than Visual-DAC due to the efficient labeling performed by the proposed STAR formulations which has detected even the noisy ground-truth images.

4.1.3. Matching accuracy based performance analysis before and after performing STAR formulation based cluster fine-tune

The matching accuracy is evaluated based on the convergence observed during the training iterations. A stable matching accuracy is essential to determine the convergence and efficiency of the labeling process. Figure 11 shows the training iteration wise matching accuracy evaluation of Visual-DAC and STAR-DAC performed on DukeMTMC re-ID dataset.

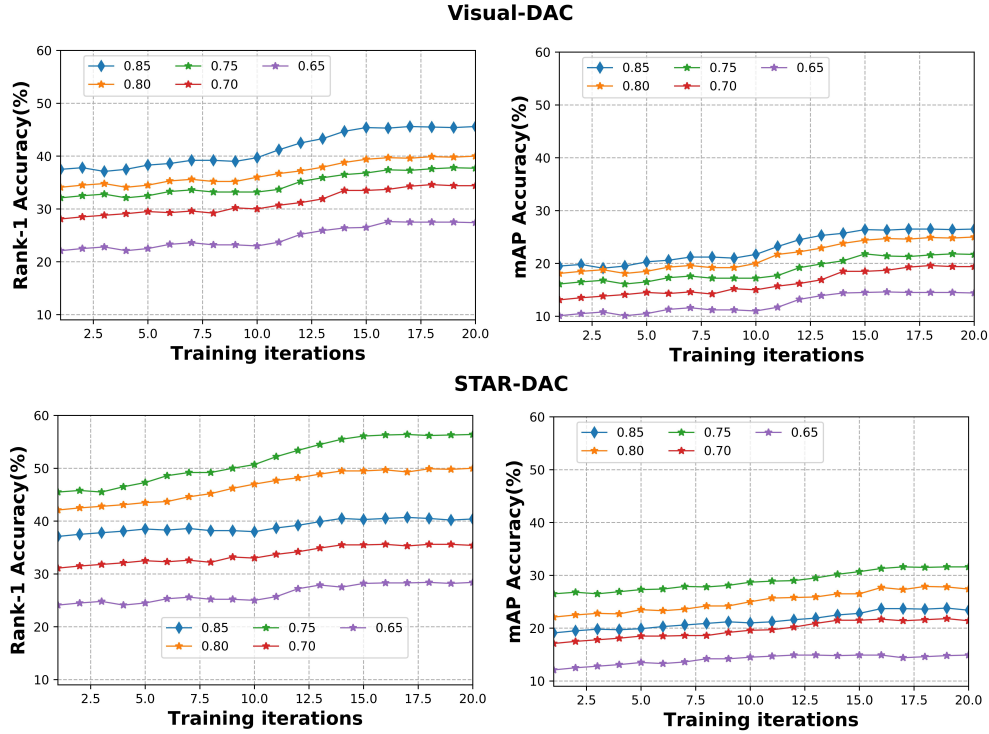


Figure 11: Analysis of the stability in matching accuracy throughout the training iterations of Visual DAC (before STAR fine-tune) and STAR-DAC (after STAR fine-tune) evaluated on DukeMTMC re-ID datasets

During the training iterations, the matching accuracy reported after performing the STAR formulations (STAR-DAC) is more stable than the baseline (Visual-DAC). Stability denotes the variation in matching accuracy reported throughout

the training iterations. High variation in matching accuracy denotes less stability and less variation denotes high stability. The effect of a high stability in STAR-DAC than Visual-DAC denotes that the matching accuracy reported by Visual-DAC possess more fluctuations and less convergence rate than STAR-DAC. The distortion in initial training iterations is due to the inability of baseline to match the noisy query images to its corresponding ground-truth identities. However, the STAR-DAC method has a more stable cluster due to efficient noisy sample labeling which in turn avoids high fluctuations in the matching accuracy during the training process.

Figure 12 and 13 show the CMC score variation for different threshold values during STAR cluster fine-tune procedure for DukeMTMC re-ID and Market-1501 dataset. The similarity threshold value δ is initialized to 0.85. Then, STAR fine-tune procedure is performed to evaluate each of the STAR formulations on the formed visual clusters. Similarity threshold value δ is gradually decreased. For each decrease in δ value, STAR formulations are evaluated to increase reliable image selection. This gradual decrease in δ is performed until there is a drop in CMC score.

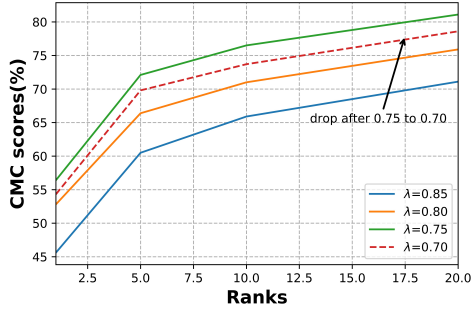


Figure 12: DukeMTMC re-ID threshold leverage analysis during STAR fine-tune

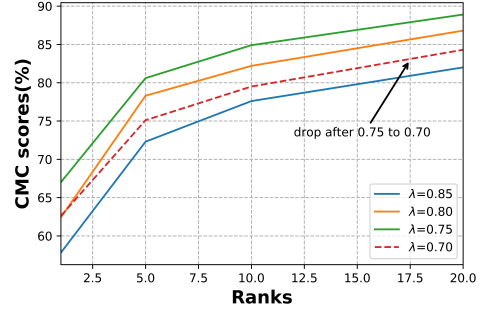


Figure 13: Market1501 threshold leverage analysis during STAR fine-tune

Reliable image selection process is stopped at the point where the CMC score drops. For a given dataset, the minimum similarity threshold value δ_{min} is concluded at this CMC drop point. Minimum similarity threshold δ_{min} value varies for different datasets based on its image variations. As observed for all the datasets, on an average the CMC score drops at δ value between 0.75 to 0.70. Thus, the δ_{min} value is finalized as 0.75 to make the process reliable.

625 Table 7 presents the characteristic comparison of existing person re-ID methods with the proposed STAR-DAC framework. The methods are compared in terms of visual and spatio-temporal features, support towards ground-truth labeling, learning techniques adopted and support of external annotations.

630 Table 7: Comparison of existing unsupervised person re-ID methods with the proposed method

Methods	Reliable sample selection	Types of data used (I,V,ST)*	Ground truth labeling	Learning Approach (U/SS/S)**	Without annotations support
UMDL [6]	x	I	x	U	x
PTGAN [7]	x	I	x	U	x
PUL [9]	✓	I	x	U,SS	x
CCLF [10]	✓	I	x	U,SS	x
PUCL [11]	✓	I	x	U,SS	x
FMC [12]	✓	I	x	U,SS	x
SSDAL [13]	x	I	x	SS	x
AGG(DIMN) [14]	x	I	x	DG	x
DualNorm [15]	x	I	x	DG	x
UnKiss [39]	x	V	x	U	x
DGM+IDE [25]	x	V	x	U	x
TAUDL [19]	x	I,V	x	U	x
UTAL [20]	x	I,V	x	U	x
SMP [24]	x	V	x	U	x
DAL [27]	x	V	x	U	x
CAMEL [22]	x	V	x	U	x
TJ-AIDL [21]	x	V	x	U	x
DAS [26]	x	V	x	U	x
stReID [4]	x	I,ST	x	S	x
TFusion [29]	x	I,V,ST	x	SS	x
Joint ReID [5]	x	I,ST	x	S	x
Temporal [30]	x	I,ST	x	SS	x
STAR-DAC	✓	I,V,ST	✓	U	✓

*I-Image data, V-Video data(Tracklets), ST-Spatio-Temporal data

**U-Unsupervised, SS-SemiSupervised, S-Supervised, DG-Domain Generalization

635 Table 7 clearly show that the literature lacks methods which incorporate spatio-temporal patterns in a complete unsupervised setting without external annotations. Thus, to the best of our knowledge, the proposed STAR-DAC framework is the first to incorporate a spatio-temporal model into the unsupervised approach of person re-ID.

640 Figure 14 show the increase in CMC score (rank-1 and mAP accuracy) of STAR-DAC than the best of the discussed literature works. The large

scale datasets and multiple camera datasets DukeMTMC re-ID, Market-1501, MSMT17, RPI field, Dukevideo re-ID and iLIDSvid reports a good performance increase than the small scale CUHK03, GRID and ViPer datasets. RPI field is not depicted since it does not have literature works to compare. GRID, iLIDSvid and ViPer has no reported mAP accuracy in the literature to compare with the proposed STAR-DAC.

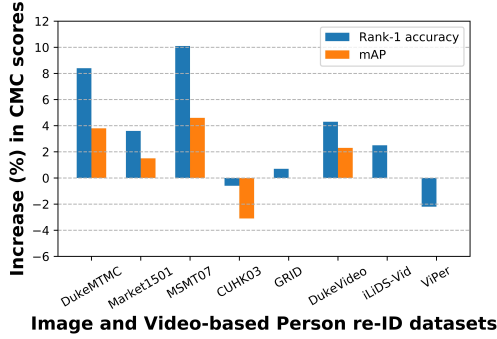


Figure 14: Accuracy improvement from Literature works to STAR-DAC

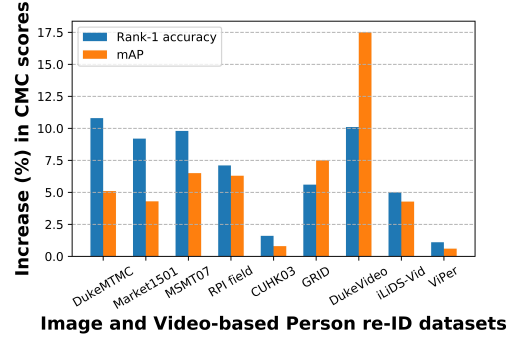


Figure 15: Accuracy improvement from Visual-DAC to STAR-DAC

Figure 15 show the CMC score increase of STAR-DAC than the Visual-DAC. The accuracy of Visual-DAC is evaluated with more number of clusters than the ground-truth identities due to similar elements of different identities merging into a cluster. The proposed STAR-DAC rectifies the drop by Visual-DAC and stabilizes the clusters through STAR fine-tune formulations. The inferences prove that large scale dataset with multiple cameras utilize the advantage of proposed STAR-DAC framework than small scale datasets.

5. Conclusion and future works

The proposed STAR-DAC framework is an unsupervised approach which labels the unlabeled person re-ID images. The spatio-temporal patterns are formulated to fine-tune the visual clustering process. STAR-DAC has reduced the issue of similar images of different identities converging into a single cluster. Outlier elimination and reliable sample selection are the two major processes which contribute in resolving the visual similarity issue. STAR formulations based cluster fine-tune process allows upto 75% of images during reliable sample selection process which leads to robust spatio-temporal pattern formation. The formed spatio-temporal patterns are formulated

into association rules to fine-tune the visual clustering process. The images
665 clustered beyond the minimum threshold value are stored as noisy images.
These noisy images are not used for visual feature formation or spatio-
temporal pattern formulations. It is used to label the complete dataset
to its ground-truth labels by leveraging the similarity threshold until all
the images are included in the process. The approach helps in eliminating
670 outliers and handling complex query images. Thus, STAR-DAC framework
takes advantage of multiple camera networks and large scale datasets. In
future, non-visual cues and pattern formation will be further explored to
improve the unsupervised person re-ID performance in a complex surveillance
environment. To explore the non-visual cues, additional domain adaptation
675 and cognitive interpretation is essential. Irrespective of the base person re-
ID method proposed, the STAR formulations can be used to fine-tune the
clusters during post ranking process.

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